

Programming Assignment #4

Very Large-Scale Rectilinear Minimum Spanning Tree

1 Problem Description

Given up to **10,000,000 points** on a 2D plane, construct the **Rectilinear Minimum Spanning Tree (RMST)** under the Manhattan distance metric. For two points $p_i = (x_i, y_i)$ and $p_j = (x_j, y_j)$, the edge weight is defined as

$$d(p_i, p_j) = |x_i - x_j| + |y_i - y_j|.$$

Your task is to compute the **total weight** of the rectilinear minimum spanning tree. Since the input size is extremely large, an $O(n^2)$ algorithm is not acceptable. Your implementation should emphasize scalability, memory efficiency, and robustness.

Your implementation should be capable of processing very large inputs efficiently while avoiding construction of the complete graph. All distance calculations and the final RMST weight must be stored using signed 64-bit integers. The program should output only the total weight of the rectilinear minimum spanning tree and must satisfy the specified time and memory limits. Solutions are expected to employ scalable geometric algorithms, such as sweep-line methods, divide-and-conquer techniques, geometric transformations combined with Kruskal's algorithm, or other provably correct approaches for large-scale rectilinear minimum spanning tree construction.

2 Input

The input describes a set of points in the plane. The first line contains the number of points n . Each of the next n lines contains the coordinates (x_i, y_i) of a point, where n is the number of points, $1 \leq n \leq 10,000,000$, x_i and y_i are integer coordinates, $-10^9 \leq x_i, y_i \leq 10^9$, and multiple points may share identical coordinates.

Input Format	Sample Input
// Number of points	5
// x_1, y_1	0 0
// x_2, y_2	2 0
// x_3, y_3	2 3
// $\vdots$	5 1
// x_n, y_n	6 4

3 Output

Output a single integer representing the total weight of the rectilinear minimum spanning tree of the given points.

Input Format	Sample Output
// Total RMST weight	13

4 Language/Platform

1. Language: C or C++ is preferred.
2. Platform: Linux.

5 Command-line Parameter

In order to test your program, you are asked to add the following command-line parameters to your program (e.g., RMST input.dat output.dat):

[executable file name] [input file name] [output file name]

6 Submission

You need to submit a tar file, which includes (1) source codes, (2) Makefile, (3) a text readme file (readme.txt) stating how to build and use your program, to E3 (<https://e3.nycu.edu.tw/>) by the deadline. Please put all required files in a folder, and use the following command to compress the folder in the linux environment.

```
tar cvf Student_ID.tar Student_ID
```

The folder name and the compressed file name must be your student ID.

7 Grading Policy

This programming assignment will be graded based on (1) the **correctness**, (2) **running time**. For each test case, it will be regarded as “failed” if your program takes more than 1 hours.